



University
of Glasgow

**Tuesday 20 May 2025
09:30-11:30 BST
Duration: 2 hours
Timed exam – fixed start time**

DEGREES of MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

Information Retrieval M COMPSCI5011

(Answer All Questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

- 1 (a) The following documents have been processed by an IR system where stemming is *not* applied:

DocID	Text
Doc1	australia is world champion 1999 australia won
Doc2	france and new zealand played each other in the semifinal
Doc3	france was in the semifinal 1999 1999
Doc4	australia won the other semifinal in wales 1999 won

- (i) Assume that the following terms are stopwords: and, in, is, the, was. Construct an inverted file for these documents, showing clearly the dictionary and posting list components. Your inverted file needs to store sufficient information for computing a simple $tf \cdot idf$ term weight, where $w_{ij} = tf_{ij} \cdot \log_2(N/df_i)$

[6]

- (ii) Compute the term weights of the two terms “champion” and “1999” in Doc1. Show your working. Note that $\log_2(0.75) = -0.4150$ and $\log_2(1.3333) = 0.4150$.

[2]

- (iii) Assuming the use of a best match ranking algorithm, rank all documents using their relevance scores for the following query:

1999 france

Show your working.

[3]

- (b) Consider a query q for which there are 10 relevant documents in a corpus of 100 documents. Consider an IR system whose ranking for this query is shown below. In particular, its top 15 ranked documents were judged for relevance where R means the document was deemed relevant and N otherwise (the leftmost item is the top ranked search result).

System ranking: R N N R R N N R N R N N N N R

- (i) Compute the precision, recall and the precision at rank 10 ($P@10$) of the system for this query. *Show your calculations.*

[3]

- (ii) Compute the average precision (AP) of the system for this query. Show your calculations.

[2]

- (c) Automatic query reformulation is a key component in many information retrieval systems. Consider two retrieval pipelines expressed in PyTerrier as follows:

```
bm25 = pt.terrier.Retriever(index, wmodel='BM25')
pipe1 = bm25 % 3 >> rewriter1(index) >> bm25
pipe2 = rewriter2 >> bm25
```

You may assume that these pipeline formulations are correct.

Assuming that the input to both pipelines is a Q dataframe, what information do rewriter1 and rewriter2 use, respectively, to rewrite the query? Name examples of possible approaches for rewriter1 and rewriter2, and describe their operation.

[4]

2. (a) Consider a corpus of documents C written in English, where the frequency distribution of words approximately follows Zipf's law $r * p(w_r|C) = 0.1$, where $r = 1, 2, \dots, n$ is the rank of a word by decreasing order of frequency. Hence, the words are ordered by decreasing order of probability of occurrence in the corpus such that w_r is the word at rank r , and $p(w_r|C)$ is the probability of occurrence of word w_r in the corpus C.

What proportion of word occurrences would be removed from the collection if we ignored all occurrences of the four most frequent words in the collection? Justify your answer.

[5]

- (b) Consider the query "jackson star" and the following term frequencies for the three documents D1, D2 and D3, where the search engine is using raw term frequency (TF) but no IDF:

	india	jackson	life	mood	pop	spice	star	zebra
D1	4	0	2	1	1	3	1	0
D2	0	4	1	0	0	1	3	1
D3	0	6	4	0	4	1	4	1

Assume that the system has returned the following ranking: D2, D3, D1. The user judges D3 to be relevant and both D1 and D2 to be non-relevant.

- (i) Show the original query vector, clearly stating the dimensions of the vector.

[2]

(ii) Use Rocchio's relevance feedback algorithm (with $\alpha=\beta=\gamma=1$) to provide a revised query vector for "jackson music". Terms in the revised query that have negative weights can be dropped, i.e. their weights can be changed back to 0. *Show all your calculations.*

[4]

- (c) We discussed a number of IR models in the class. These models can make a number of assumptions to simplify the relevance estimation of a document given a query. Provide 4 examples of assumptions made by IR models, which do not necessarily hold true. In particular, for each assumption, briefly provide a concrete and informative example showing why the assumption does not hold in a real-world search environment.

[4]

- (d) Consider a query with two terms, whose posting lists are as follows:

term1 $\rightarrow$ [id=2, tf=2], [id=5, tf=1], [id=6, tf=2]

term2 $\rightarrow$ [id=2, tf=4], [id=3, tf=5], [id=5, tf=5]

Explain and provide the exact steps/order in which the posting lists will be traversed by the TAAT & DAAT query evaluation strategies and the memory requirements of both strategies for obtaining a result set of K documents from a corpus of N documents ($K < N$).

[5]

3. (a) List four key components shared by most advanced retrieval models (such as BM25 or PL2) we have learnt in the class. Briefly describe why they are important in an IR system.

[4]

- (b) Consider a situation where you are building a search engine for products on an e-commerce website. You start with a basic lexical search system using the BM25 weighting model, along with Porter stemming and stopword removal.

(i) Users report that when they search for "i-phone" they do not get the results they are looking for. Using what you know about tokenization, why is this happening? Propose one solution to fix this problem that still uses lexical search.

[2]

(ii) Your company earns more money on some brands than others. Your CEO wants you to improve the ranking of items that earn more money when multiple products are relevant. What changes would you need to make to a Learning-to-Ranking system to facilitate this?

[2]

(iii) Hybrid search engines combine both lexical and dense retrieval scores. In the context of an e-commerce website, provide one advantage of lexical retrieval, one advantage of dense retrieval, and one advantage of combining the signals.

[3]

(iv) The hybrid search engine applies a cross-encoder reranker to improve the ranks of the relevant documents. The top 10 documents are presented to the user. Assuming you have PyTerrier transformers named *lexical*, *dense*, and *reranker*, propose a pipeline formulation for the hybrid search engine. You may assume that the *reranker* is applied to all documents retrieved by the first-stage retriever.

[2]

(v) Your CEO hypothesises that the hybrid search engine will improve results for users compared with using only a sparse or dense as the first-stage ranker.

Propose two experiments that would verify this hypothesis. You may use PyTerrier-esque experiment notation. Justify clearly your choice of pipelines, and your choice of evaluation measures.

You may assume access to a test collection through *topics* and *qrels* variables in your experiments.

[7]